Marmara Üniversitesi
Electrical and Electronics Engineering Department
EE 3011: Electronics I (Spring 2024)

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Course Information	Text: Sedra/Smith, *Microelectronic Circuits 5th Ed.*	Grading	Class Work 15%	Lab Work 15%	Mid-term Exam 30%	Final Exam 40%

Week	Events	Topics to cover	Ch	Sec	Lab Activity	Exercises
1 Feb 12-16	---	**Introduction to Physics of Semiconductors.** Conduction in Semiconductors, N-type and P-type Semiconductors, The PN junction.	3	7	NL	(**3**) 112,113,114, 116,118.
2 Feb 19-23	---	**Diodes**. Introduction, The Ideal diode, Terminal characteristics of the Diode.	3	1-2	Experiment (1) LTSpice	(**3**) 2,3,4,6,9,10.
3 Feb 26-Mar 1	Course drop/add Week	**Diode Circuit Analysis**. Diode Models, Graphical and Analytical Diode Circuit Analysis.	3	3	Experiment (2) LTSpice	(**3**) 17,20,22,25,26, 40,41,46,47.
4 Mar 4-8	---	**Diode Applications**. The Zener diode. Half and Full-wave Rectifiers.	3	4-6	Experiment (3) Diode ckt Simulation	(**3**) 64,65,66,71,76, 78,82.
5 Mar 11-15	Quiz 1 March 12 (In-class)	**Diode Applications**. Limiting and Clamping Circuits, Voltage Doublers. Other Diode types. Examples	3	8	Experiment (4) Basic Diode Circuits	(**3**) 93,98,105.
6 Mar 18-22	---	**MOS Field-Effect Transistors (MOSFETs).** Device Structure and Modes of Operation. Types, Symbols and Conventions.	4	1-2	Experiment (5) Diode Rectifiers	(**4**) 4,6,12,15,19,23.
7 Mar 25-29	---	**MOSFET DC Circuits.** Biasing and Circuit DC Analysis.	4	3	Problem Session	(**4**) 35,36,37,41,42, 44,47.
April 1-7	**Midterm Exams period**					
		Apr 9-12 Holidays				
9 Apr 15-19	---	**MOSFET Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CS, CG and CD).	4	4-6	NL	(**4**) 63,69,74,75,77.
10 Apr 22-26	Midterm Make-up (Excused, Week 11)	**MOSFET Amplifiers.** Single-stage Amplifiers (CS, CG and CD).	4	7	Experiment (6) MOSFET DC Ckts	(**4**) 81,85,88.
11 Apr 29-May 3	Labor Day: May 1	**Bipolar Junction Transistors (BJTs).** Structure and Modes of Operation, Types, Symbols and Conventions,	5	1-2	NL	(**5**) 3,4,7,13,20, 31,39.
12 May 6-10	---	**BJT DC Circuits**. Biasing. Current-voltage (IV) Characteristics. BJT as a switch.	5	4	Experiment (7) MOSFET Amplifiers	(**5**) 69,71,72,74, 79,82,84,86.
13 May 13-17	Quiz 2 May 14 (In-class)	**BJT Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CE, CB and CC).	5	6	Experiment (8) BJT Circuits	(**5**) 104,106,107, 115,116,124.
14 May 20-24	YS Day May 19	**BJT Amplifiers.** Single-stage Amplifiers (CE, CB and CC).	5	7	NL	(**5**) 130,131,134, 137,141,143.
15 May 27-31	---	**CMOS Logic Circuits.** Overview, the CMOS Inverter, Other CMOS Logic circuits, CMOS transistor sizing	10	1-3	Experiment (9) BJT Amplifiers	(**10**) 18,25,28.
June 3-14	**Final Exams period**	**July 1-7 (Re-sit Final Exams period)**				